

# Day 1 — AI Spec Development

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**Activity packet** for participant quads. Today's job: install the **AI Spec Development pattern** — a structured prompt sequence that produces an engineering-ready integrated technical spec from PRD/TCD/TMD inputs — and lock the **Holocron scope slice** you will specify all week.

## Where we are in the week

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Week 8 opens with a new technique. AI Spec Development is the synthesis of everything learned across Weeks 2 (prompting), 4–5 (architecture and modeling), and Week 5 Day 5 (validation discipline). The pattern is what an experienced TPM uses to compress a week of spec work into hours — without losing the rigor.

By 16:00, every quad has:

- A **working AI Spec template** with the prompt sequence
- A **locked Holocron scope slice** with initial inputs gathered
- **Tomorrow's plan** — what discovery they need to do for compressed PRD work

## Inputs

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- The full sibling artifact set from Weeks 3–7 (the templates carry forward)
  - The **Holocron problem brief** (issued end of Week 7)
  - The Pattern Library accumulated across Weeks 2, 4, 5, 6, 7
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## What "AI Spec Development" is — and isn't

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| Is                                                                          | Is not                                          |
|-----------------------------------------------------------------------------|-------------------------------------------------|
| A structured sequence of validated prompts that produces an integrated spec | "Just paste everything and ask AI to summarize" |
| Anchored on the PRD/TCD/TMD/SEP/DP outputs as inputs                        | Replacing the artifact set                      |
| Disciplined: every output is validated against a source                     | Trust the AI's output                           |
| Iterative: 4–6 prompts in sequence, each refining                           | One mega-prompt                                 |
| Engineering-ready output — sections an engineer can scope from              | A pretty draft for stakeholders                 |
| Always paired with a provenance log                                         | Untracked AI use                                |

A senior TPM uses this pattern to **draft** the spec; their judgment shapes every section; AI accelerates the typing, not the thinking.

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## The AI Spec template (today's reference)

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# AI Spec – <feature>
**Sibling to:** PRD <link>, TCD <link>, TMD <link>,
SEP <link>, DP <link>
**Authors:** <quad> | **Status:** Draft / Reviewed
**AI Provenance:** see end-of-doc log

## 1. Headline
One sentence: what is this feature, who is it for,
what outcome does it produce?

## 2. Engineering-ready summary
3–5 paragraphs that an engineer could scope from
without a clarifying call.
Synthesizes: PRD Sections 1–5, TCD Section 1, TMD
Section 3.

## 3. Data + API contract
The shape an engineer needs to start coding.
Synthesizes:
TMD Section 1 (data) + TMD Section 3 (API).

## 4. Sequence + failure handling
Happy / sad / weird path summary. Synthesizes: TMD
Section 4.

## 5. Constraints
Performance / availability / security / compliance /
accessibility.
Synthesizes: TCD Section 3 + TCD Section 4 + PRD
Section 7.

## 6. Decisions made (and not made)
Top trade-offs (TCD Section 5) + open questions
still requiring decision.

## 7. Stakeholders + sign-off
Who must sign off on what. From SEP Section 1 + TCD
Section 6.

## 8. Provenance log
Every AI prompt + validation status.
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The AI Spec is **shorter** than the sum of its inputs. It's the synthesis an engineer would read first, with citations back to the deeper artifacts.

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## The 5-prompt sequence (today's main pattern)

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The AI Spec is generated from a **5-prompt sequence**, not a single prompt. Each prompt has a specific job + validation step.

### Prompt 1: Headline + engineering-ready summary

Role: Senior TPM drafting an engineering-ready spec.  
Context: Below are excerpts from PRD Sections 1–5, TCD Section 1, TMD Section 3.  
<paste excerpts>  
Task: Produce:

1. A one-sentence headline (what / who / outcome)
2. 3–5 paragraphs that an engineer could scope from without a clarifying call. Synthesize the inputs; do not invent.

Constraints:

- Use only the provided content
- Cite sections you draw from
- Flag anything ambiguous in the input

Format: Headline + numbered paragraphs + footnotes section.

**Validation:** read through; verify every claim cites a section. Cut hallucinated claims.

### Prompt 2: Data + API contract synthesis

Continuing. Below are TMD Section 1 (data) and TMD Section 3 (API).  
<paste>  
Task: Produce a 1-page synthesis that an engineer could implement against. Include:

- Entity-level data shape (key fields + relationships)
- API surface (endpoints + methods + key behaviors)
- Idempotency + versioning + error semantics

Constraints:

- Do not invent fields or endpoints
- Flag any inconsistency between the data model and API

Format: Markdown with sub-sections.

**Validation:** for every entity / endpoint, verify it exists in the source.

### Prompt 3: Sequence + failure handling

Continuing. Below are the 3 sequence diagrams (happy/sad/weird) from TMD Section 4.  
<paste descriptions>  
Task: Produce:  
1. Happy-path narrative (1 paragraph)  
2. Top 2 sad-path scenarios with system response  
3. Top 1 weird-path scenario with named invariant  
Constraints:  
- Reference the sequence diagrams; do not invent flows  
- Preserve the named invariant from the weird path  
Format: Markdown with sub-headings.

**Validation:** the named invariant from TMD Section 4 is preserved verbatim.

### Prompt 4: Constraints synthesis

Continuing. Below are TCD Section 3 (security/compliance), TCD Section 4 (SLOs), PRD Section 7 (NFRs).  
<paste>  
Task: Produce a 1-page constraints summary covering:  
- Performance + availability targets (SLOs)  
- Security + compliance non-negotiables  
- Accessibility floor  
- Observability requirements  
Constraints:  
- Use the actual numbers / targets from the inputs  
- Flag any constraint that conflicts with another  
Format: Markdown table; include "trade-off" column for any conflicts.

**Validation:** every number is traceable to a source.

## Prompt 5: Decisions + stakeholders

Continuing. Below are TCD Section 5 (top trade-offs), TCD Section 6 (sign-off matrix), SEP Section 1 (stakeholder map).

<paste>

Task: Produce:

1. The top 5 trade-offs (Option A / B / Choice / Cost / Revisit)
2. Open decisions still required (with named owner)
3. The sign-off matrix linking constraints to stakeholders

Constraints:

- Preserve the trade-off structure exactly
- Do not soften "accepted cost" or "revisit trigger"

Format: 3 sub-sections.

**Validation:** the trade-off costs and revisit triggers are preserved verbatim — these are the senior-author signals.

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## Activity 1 — Holocron Scoping

**Format:** Quad • 35 min • Block 1

### Purpose

Lock a defensible slice of the Holocron problem. The subject is fixed; what varies is how much you take on — and an over-broad slice won't survive contact with a 4-day build.

### Quad protocol

1. **Re-read the problem brief** (5 min). Compare it against the one-sentence problem you captured Friday.
2. **The 5 scoping questions** (15 min):
  - **One-sentence problem?** Can you state it so an engineer could scope from it — not a restatement of the mandate?
  - **Who is it for first?** Have you named the primary user, and can you defend choosing them over the others in the brief?
  - **Substantive but bounded?** Rich enough to carry a real spec, small enough to finish in 4 days?

- **Evidence, not invention?** Can every claim point to a fact in the brief — or be flagged as an assumption with an owner?
  - Can we **ship the artifact set in 4 days** (Day 2-4 build + Day 5 present)?
3. **Draw the scope line** (10 min). Name what is in release one and what you are deliberately deferring. If 1+ question is "no", narrow the slice.
  4. **List inputs you'll need** (5 min). Which stakeholders you'd interview, which brief facts you'll build on, which open questions you must resolve or route.

## Output

A one-sentence problem statement, a named primary user, an explicit in/out scope line, and a list of inputs to gather before tomorrow's discovery work.

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## Activity 2 — Walk Through the AI Spec Template

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**Format:** Quad • 40 min • Block 2

### Purpose

Internalize the AI Spec template by walking through it section-by-section using a **previously-completed FieldPulse artifact set** as the source.

### Quad protocol

1. **Read the AI Spec template** (5 min).
2. **Walk each section** (30 min). For each of the 8 sections, identify:
  - What input artifact does this synthesize?
  - What's likely to be missing if the input is thin?
  - What would an engineer expect here that we should make sure to include?
3. **Identify the section your team will struggle with most** (5 min). Plan a counter-measure.

## What "good" looks like

- Every section is mapped back to **specific input artifacts**
  - The "missing-if-thin" awareness is **specific** ("if the data model has no relationships diagrammed, Section 3 will be hard to draft")
  - The "section we'll struggle with" is **named** with a counter-measure
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## Activity 3 — Run the 5-Prompt Sequence on FieldPulse

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**Format:** Quad • 40 min • Block 3

### Purpose

Practice the sequence on the FieldPulse artifact set (where the inputs are complete and validated) before applying it to the capstone tomorrow.

### Quad protocol

1. **Run Prompts 1–5 in sequence** (30 min). For each:
  - Paste the relevant input
  - Capture the output
  - Validate (cross-check / spot-check / mark)
2. **Assemble the AI Spec** (10 min). Combine the 5 outputs into the Section 1–Section 7 structure. Add the provenance log.
3. **Cross-check the result against the original artifacts** (no extra time block — done as you go). Anything in the AI Spec that contradicts a source: flag and fix.

### Output

A first AI Spec for FieldPulse — produced via the 5-prompt sequence — that the quad can use as a template for tomorrow's compressed work.

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## Activity 4 — Capstone Discovery Plan

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**Format:** Quad • 45 min + Wrap • Block 4



## Purpose

Plan the discovery work needed for tomorrow's compressed PRD.  
Distribute the work across quad members.

## Quad protocol

1. **List discovery inputs needed** (10 min). For your capstone:
  - Customer signal — interviews, support data, public reviews
  - Strategic context — why this matters, what's been tried
  - System inputs — existing docs, codebase access, related artifacts
  - Stakeholder signal — who has opinions; what are they
2. **Distribute work** (10 min). Each quad member takes 1–2 categories.
3. **Set the deadline for inputs gathered** (5 min). Typically: 8 PM tonight or first hour tomorrow morning.
4. **Identify the "we don't have this and can't get it"** (10 min). Be honest. Document what you're proceeding without; mark it for AI-augmented hypothesis tomorrow (with extra validation).
5. **Run the AI prompt** (10 min):

```
Role: Capstone advisor for a TPM cohort.  
Context: <Holocron scope slice + 5-question scoping  
assessment +  
declared inputs needed>  
Task: Identify 3 risks in the capstone plan:  
1. A customer signal source likely insufficient  
2. A scope dimension likely too ambitious for 4  
days  
3. A stakeholder whose buy-in we'll need but  
didn't list  
Constraints:  
- Be specific to the subject given  
- Suggest a concrete adjustment per risk  
Format: 3 numbered findings.
```

## Wrap (last 15 min)

Each quad shares:

- Their **Holocron problem statement + scope line** in one sentence
  - The **section of the AI Spec they think they'll struggle with most**
  - One **risk** the AI surfaced about their plan
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## End-of-day checkpoint

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Each quad ends Day 1 with:

- ✓ **Capstone subject confirmed** through 5-question fit assessment
- ✓ **AI Spec template** internalized; FieldPulse practice run completed
- ✓ **5-prompt sequence** rehearsed
- ✓ **Discovery inputs list** + distributed assignments
- ✓ **Risks** acknowledged with mitigations
- ✓ AI provenance log entry